

Algebraic Expressions & Quadratic Equation

Covering

- Introduction to algebra
- Algebraic manipulation
- Algebraic fractions
- Equations

O levels and Extended

Simplifying Expressions

To simplify expressions we take like/common terms. Like terms are such with same variables.

E.g $2x$ and $11x$ are like but $2xz$ and $13xy$ aren't as they don't have same variables.

Expansion

We can expand expressions by recalling the distributive property, stating: $a(b \pm c) = ab \pm ac$.

To expand equations in the form $(a + b)(c + d)$, we use the FOIL method;

First	$(a+b)(c+d)$	ac
Outside	$(a+b)(c+d)$	ad
Inside	$(a+b)(c+d)$	bc
Last	$(a+b)(c+d)$	bd

$$(a + b)(c + d) = ac + ad + bc + bd$$

E.g $(2x + 3)(x + y) = 2x^2 + 2xy + 3x + 3y$

For special cases like $(a + b)^2$, $(a - b)^2$, or $(a - b)(a + b)$ we use the identities:

Square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$
Square of a difference	$(a - b)^2 = a^2 - 2ab + b^2$
Difference of 2 squares	$(a - b)(a + b) = a^2 - b^2$

E.g $(4u + x)^2 = (4u)^2 + 2(4u)(x) + (x)^2 = 16u^2 + 8ux + x^2$
 $(6m - 2l)^2 = (6m)^2 - 2(6m)(2l) + (2l)^2 = 36m^2 - 24ml + 4l^2$
 $(1 - z)(1 + z) = (1)^2 - (z)^2 = 1 - z^2$



Factorization

To factorize, we take common things from pairs or group of terms.

1. Taking a constant number common (e.g 3, 8, 99) means, taking the largest common factor from the pair/group of terms.

E.g $2x - 6y$ has 2 factor common; $2x - 6y = 2(x) - 2(3y) = 2(x - 3y)$

$18 - 3z$ has 3 factor common; $18 - 3z = 3(6) - 3(z) = 3(6 - z)$

2. Taking a variable term common (e.g x , $3z$, $2uv$) means, taking the smallest power of the common variable, dividing the terms by that common variable, then grouping it in a bracket with that term outside it. Use indices to work out complex or large powers.

E.g $2x^2 + 8xy$ has $2x$ factor common; $2x^2 + 8xy = 2x(x) + 2x(4y) = 2x(x + 4y)$

$15um - 45lm$ has $15m$ factor common; $15um - 45lm = 15m(u) - 15m(3l)$
 $= 15m(u - 3l)$

(O levels and Extended)

3. Factorization can also be done by using the 3 identities mentioned but in reverse. By finding them in a similar form to the right hand side we can factorize it into the left hand side.

E.g $4f^2 + 12fg + g^2 = (2f)^2 + 2(2f)(g) + (g)^2 = (2f + g)^2$

$81d^2 - 54db + 9b^2 = (9d)^2 - 2(9d)(3b) + (3b)^2 = (9d - 3b)^2$

$t^2 - 4 = (t)^2 - (2)^2 = (t - 2)(t + 2)$

(O levels and Extended)

4. Factorization of a 4 term expression is by: (1). making 2 pairs, (2). taking common from them. Make sure that the factors inside the brackets are same. (3). Then finally the first term is the common brackets, second term is the expression made by the common factors ignoring the brackets.

E.g $15xz + 4y - 10z - 6xy$

 1. $15xz - 10z - 6yx + 4y$

*** *****
 2. $5z(3x - 2) - 2y(3x - 2)$

3. $(3x - 2)(5z - 2y)$



Quadratic Equation**O levels and Extended**

If we have a 3-term equation of the general form $ax^2 + bx + c$, then to factorize it we have various ways:

Mid-Term breaking**O levels and Extended**

For $ax^2 + bx + c$, (1). we prime factorize ac and check to see what 2 numbers, when multiplied give us ac , and when added give us b . (2). After finding we break b into the 2 numbers we find, e.g $bx \rightarrow mx + nx$, then we use the 4 term factorization method to factorize it.

E.g $3x^2 - x - 10$.

1. We need to find 2 numbers such $m + n$ is negative, which when listing the prime factors of -30 we formulate $5 - 6 = -1$ (b) and $5 \times (-6) = -30$ (ac)

2. $3x^2 + 5x - 6x - 10 \rightarrow x(3x + 5) - 2(3x + 5) \rightarrow (3x + 5)(x - 2)$

E.g $5x^2 + 47x + 18$.

1. We need to find 2 numbers such $m + n$ is positive, which when listing the prime factors of 90 we formulate $2 + 45 = 47$ and $2 \times 45 = 90$

2. $5x^2 + 2x + 45x + 18 \rightarrow x(5x + 2) + 9(5x + 2) \rightarrow (5x + 2)(x + 9)$

Solving the Quadratic equation**O levels and Extended**

For this we can either, set it to 0, $ax^2 + bx + c = 0$, factorize and set each factor to 0 and solve the linear equations. However, if factorization cannot be done, we can instead use the Quadratic Formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We can plug in the values, a being x^2 's coefficient, b being x 's coefficient, and c being the constant term.



Completing the Square**O levels and Extended**

This is the derivation for questions requiring us to express the answer in the form

$$a(x + b)^2 + c.$$

$$ax^2 + bx + c = 0$$

$$3x^2 + 20x - 7 = 0$$

We can factor out for a in the first 2 terms

$$a \left(x^2 + x \frac{b}{a} \right) + c = 0$$

$$3 \left(x^2 + \frac{20}{3}x \right) - 7 = 0$$

We add and subtract the square of half of the coefficient of " x "

$$a \left(x^2 + x \frac{b}{a} + \left(\frac{b}{2a} \right)^2 - \left(\frac{b}{2a} \right)^2 \right) + c = 0 \quad 3 \left(x^2 + \frac{20}{3}x + \left(\frac{10}{3} \right)^2 - \left(\frac{10}{3} \right)^2 \right) - 7 = 0$$

We pull out the fourth term and factorize the first 3 terms

$$a \left(x + \frac{b}{2a} \right)^2 - a \left(\frac{b}{2a} \right)^2 + c = 0$$

$$3 \left(x + \frac{10}{3} \right)^2 - 3 \left(\frac{10}{3} \right)^2 - 7 = 0$$

Finally we simplify $\rightarrow$

$$3 \left(x + \frac{10}{3} \right)^2 - \frac{121}{3} = 0$$

We can go further and prove the quadratic formula, i.e solve for x .

$$a \left(x + \frac{b}{2a} \right)^2 - a \left(\frac{b}{2a} \right)^2 + c = 0 \quad 3 \left(x + \frac{10}{3} \right)^2 - \frac{121}{3} = 0$$

Shift the constant to the RHS

$$a \left(x + \frac{b}{2a} \right)^2 = a \left(\frac{b}{2a} \right)^2 - c \quad 3 \left(x + \frac{10}{3} \right)^2 = \frac{121}{3}$$

divide by a , simplify

$$\left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a^2} \quad \left(x + \frac{10}{3} \right)^2 = \frac{121}{9}$$

take $\sqrt{\quad}$, remember to put $\pm$

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \quad x + \frac{10}{3} = \pm \frac{\sqrt{121}}{3}$$

isolate x

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad x = \frac{-10 \pm 11}{3}$$

$$x = 7, \quad \frac{1}{3}$$



Algebraic Fractions**O levels and Extended**

We basically combine fractions how we would from Numbers, except with algebraic expressions. Multiply the denominator of the other fraction on both sides, then combine. If there is a $-$ then bracket the denominator of the second fraction.

E.g $\frac{x}{3} + \frac{x-4}{2}, \quad \frac{3}{x+5} - \frac{2}{x}$

$$\frac{x}{3} + \frac{x-4}{2}$$

$$\frac{3}{x+5} - \frac{2}{x}$$

$$\frac{x \times 2}{3 \times 2} + \frac{(x-4) \times 3}{2 \times 3} \quad \frac{3 \times x}{(x+5) \times x} - \frac{2 \times (x+5)}{x \times (x+5)}$$

$$\frac{2x + 3x - 12}{6}$$

$$\frac{3x - (2x + 10)}{x^2 + 5x}$$

$$\frac{5x - 12}{6}$$

$$\frac{x - 10}{x^2 + 5x}$$

To deal with fractions with 2 expressions, with 1 or 2 quadratic equations, we just factorize them and remove the common factors, leaving us the uncommon factors.

E.g $\frac{x^2 - 2x}{x^2 - 5x + 6} \rightarrow \frac{x(x-2)}{(x+3)(x-2)} \rightarrow \frac{x}{(x+3)}$

Subject of formula

For this we rely on Additive and Multiplicative Identities, stating $a - a = 0$ and $a \times \frac{1}{a} = 1$. We shift around terms, by applying operations on both sides, to isolate the wanted variable. If we have the variable more than 1 places, then we group all terms with it to one side, group and divide by the coefficient of the subject variable.

E.g $\frac{d}{1-r} = 2$, making r subject of formula is to: $\times \left(\frac{1-r}{2}\right), \times -1, +1$ on both sides to

get $r = 1 - \frac{d}{2}$



Word Problems**O levels and Extended**

When making equations from statements we make/assume unknown values x and add or subtract according to the wordings given. Make sure x and other things are in the same unit. Make equations with given conditions, a hint can be the question wanting the derived answer.

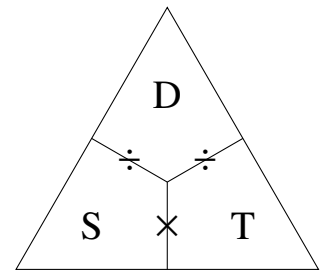
For speed related problems, the speed triangle:

$$(\text{km, m}) \quad \text{Distance} = \text{Speed} \times \text{Time}$$

$$(\text{km/h, m/sec}) \quad \text{Speed} = \frac{\text{Distance}}{\text{Time}}, \quad \text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

$$(\text{h, sec}) \quad \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{km/h} \times \frac{5}{18} = \text{m/sec}, \quad \text{m/sec} \times \frac{18}{5} = \text{km/h}$$

**Simultaneous Equations**

For this we can employ Substitution method, 1. solve for one of the variables in either equation, 2. then substitute the first into second or vice versa simple linear equation. After solving for that variable, 3. substitute it in back into either equation and solve for the other variable. For Extended, this works for both linear and non-linear cases given

E.g $2x + 3y = 15$ (1) I will solve for x in (1) and y in (2).

$$-x + 5y = 2 \quad (2)$$

$$1. \quad \begin{aligned} 2x + 3y &= 15 \\ x &= \frac{15 - 3y}{2} \end{aligned}$$

$$\begin{aligned} -x + 5y &= 2 \\ y &= \frac{2 + x}{5} \end{aligned}$$

↓

$$2. \quad \begin{aligned} -x + 5y &= 2 \\ -\left(\frac{15 - 3y}{2}\right) + 5y &= 2 \\ y &= \frac{19}{13} \end{aligned}$$

$$\begin{aligned} 2x + 3y &= 15 \\ 2x + 3\left(\frac{2 + x}{5}\right) &= 15 \\ x &= \frac{69}{13} \end{aligned}$$

$$3. \quad x = \frac{15 - 3\left(\frac{19}{13}\right)}{2} = \frac{69}{13}$$

$$y = \frac{2 + \frac{69}{13}}{5} = \frac{19}{13}$$



Inequalities

Covering

- Inequalities

Inequalities

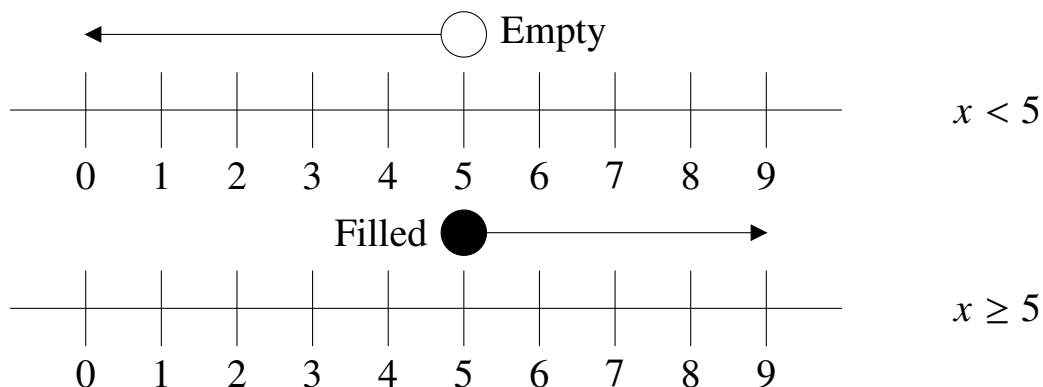
For this the symbols $<$, $\leq$, $\geq$, $>$ are used to indicate what values satisfy the original inequality.

$<$, $>$ represent values less than, or greater than, the right side, but not including that value. E.g $x < 5$ means values less than 5, but not 5. However, $\leq$, $\geq$ do include the value. E.g $x \leq 5$ means less than or equal to 5, 5 is included.

Inequalities on Line

We can represent inequalities on a number line.

For $<$, $>$ we use a empty circle, and a arrow towards the direction of the inequality symbol.



Compound Inequality

Ones with 3 equations, the middle one is taken as common and cut it into 2 inequalities. One with the first and common(middle), one with the common(middle) and third.

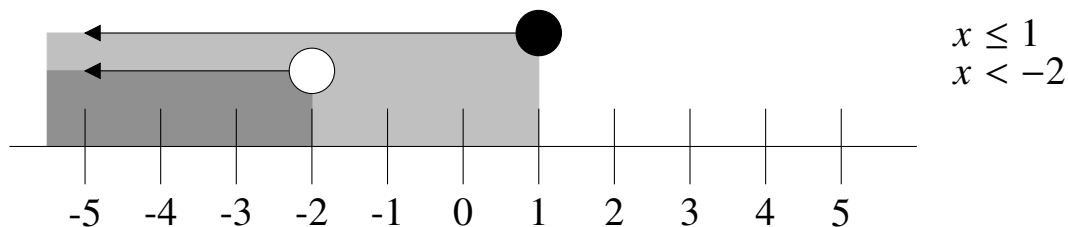
E.g

$$\begin{array}{l} \text{———— } 9x - 5 \leq 3x + 1 < x - 3 \text{ ————} \\ 9x - 5 \leq 3x + 1 \qquad \qquad \qquad 3x + 1 < x - 3 \\ 6x \leq 6 \qquad \qquad \qquad 2x < -4 \\ x \leq 1 \qquad \qquad \qquad x < -2 \end{array}$$



Solutions of Inequalities

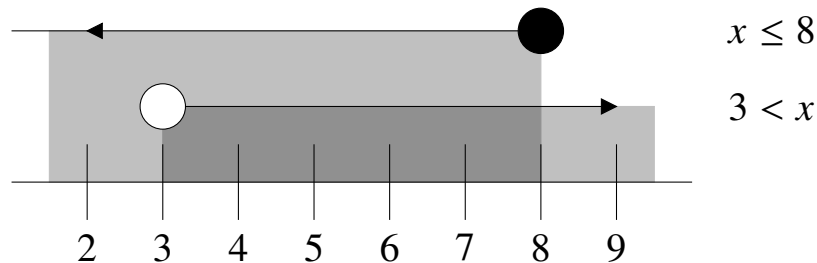
The solution of compound inequalities is either 2 arrows which don't overlap, or which do overlap. E.g for the example inequality $9x - 5 \leq 3x + 1 < x - 3$ it is $x \leq 1$, $x < -2$



Here solutions overlap but we see the overlapping region is $x < -2$, so the final answer is $x < -2$. The shaded region also shows this.

Questions which have 2 inequalities with "and" in the middle, the solution(s) of either inequality must satisfy the other. Aka, only the common solution should be considered.

E.g if we have; Solve the inequalities $3x + 4 > 13$ and $3x - 13 \leq 11$. We can derive the answers $3 < x$, $x \leq 8$. When placed on the number line, the over



Then the solutions are $3 < x \leq 8$ as they overlap to create a little shaded region.

Shaded Inequalities

To write inequalities of shaded regions.

Write $\leq$ for regions below given line on the graph or left of given vertical line. For a dotted line or if we have "inside" use $<$ instead.

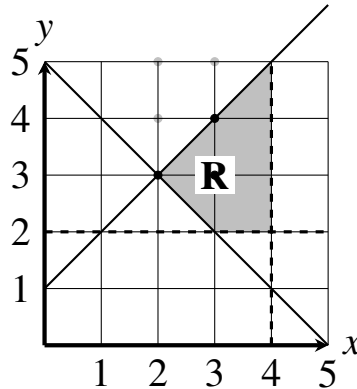
Write $\geq$ for regions above given line on the graph or right of given vertical line. For a dotted line or if we have "inside" use $>$ instead.

These rule apply to shading required regions.



Shaded Inequalities

E.g



For this we have 4 lines which define the shaded region; $x + y = 5$, $y - x = 1$, $y = 2$, $x = 4$. To get the shaded region we do:

$\leq$; below line	$x + y \leq 5$
$\geq$; above line	$y - x \geq 1$
$>$; above line + dotted	$y > 2$
$<$; left of line + dotted	$x < 4$

Line equations are either given or need to find them manually.

Inequality Arithmetic

When dividing/multiplying by a negative number, flip the sign.

E.g to solve $-x < 3$, multiply/divide by -1 but flip the sign, $x > -3$, and fix it $-3 < x$.

Possible Values

When asks for possible integers, solve for x then plug in values within the bounds.

When asks for largest or smallest value, solve for the variable and get the largest or smallest possible value which satisfy equation.

When asks for possible coordinates, consider all possible given conditions. To find the ranges of value to test, look at the left most vertex to right most vertex of the region to determine the x bounds. Look at the bottom most vertex to top most vertex of the region to determine the y bounds.

E.g possible values of (x, y) which lie inside in the shaded region in the above graph can be $2 \leq x < 4$ for x , $2 < y \leq 5$ for y . Plot them and see which ones lie inside: $(2, 3)$, $(3, 3)$, $(3, 4)$



Sequence and Pattern

Covering

- Sequences

Sequence and Pattern

A number sequence is a set of numbers arranged to a specific rule. The numbers in the sequence are known as terms of sequence.

Question types

Write missing term in sequence

For this we just continue the pattern, e.g if its adding 2, multiplying by 5, etc. Or check by the type of numbers, e.g squares or cubes.

E.g Write the next 2 terms of each;

- (a) 1, 4, 9, 16, ...: 25, 36; Squares
- (b) 1, 1, 2, 3, 5, 8, ...: 13, 21; Fibonacci Sequence
- (c) 1, 8, 27, 64, ...: 125, 216; Cubes
- (d) 5, 3, 1, -1, ...: -3, -5; Arithmetic Sequence; rule -2
- (e) 2, 3, 5, 7, 11, ...: 13, 17; Prime numbers

Complete the table

To complete the table, check the sequence of each row, or column, or by help of the diagram and follow the pattern shown.

E.g a table may be like this.

1	2	3	4	5	...	n	
5	8	11	14	p	...	x	$p = 14 + 3 = 17$
4	9	16	25	q	...	y	$q = 5^2 = 25$
1	8	27	64	r	...	z	$r = 5^3 = 125$
10	25	53	103	s	...	t	$s = 17 + 25 + 125 = 179$



Write specific term of given n^{th} term

The n^{th} term of a sequence is the formula/equation of a sequence evaluated at the wanted value(s). Plug in the wanted Term of the sequence in the formula which is in terms of n .

E.g The n^{th} term of this sequence is $57 - 2^n$. Calculate the 5^{th} term.

$$T_5 = 57 - 2^5 = 57 - 32 = 25$$

Write n^{th} term of a given sequence

There are three sequence types;

Arithmetic Sequence

If the difference between terms is constant, it calls for Arithmetic Sequence.

$T_n = a + d(n - 1)$ is used to find the n^{th} term. a is the first term (T_1), d is difference ($T_n - T_{n-1}$), n is the term number.

E.g A sequence is 7, 4, 1, -2, -5, n^{th} term or T_n is:

$$d = T_2 - T_1 = 4 - 7 = -3, \quad T_n = 4 - 3(n - 1) = 7 - 3n$$

Square or Cube Sequences

When difference between terms isn't constant, and it follows a square or cube patterns:

E.g 1, 4, 9, 16, The n^{th} term is $T_n = n^2$

E.g 1, 8, 27, 64, The n^{th} term is $T_n = n^3$

Related Sequences

When you are given a sequence with it's n^{th} term, and you want to find the n^{th} term of another related sequence which is obtained by adding, subtracting, multiplying or dividing the sequence by a constant, then perform the same operations on the n^{th} term.

E.g For sequence 4, 9, 16, 25, ..., we see it's $T_n = n^2$ but with n^{th} term 1 more, hence $T_n = (n + 1)^2$.

E.g For sequence 1, 3, 6, 10, 15, ... we notice it's the sum of positive integers unto n , aka the triangle numbers. $T_n = \frac{1}{2}n(n + 1)$ (remember this, useful)



Related Sequences

E.g Sequence 3, 5, 8, 12, 17, ... if we subtract by 2 we get the triangle numbers, meaning the sequence is just $T_n = \frac{1}{2}n(n+1) + 2$

E.g Sequence 3, 9, 18, 30, 45, ... if we divide by 3, we get triangle numbers, meaning $T_n = 3 \times \frac{1}{2}n(n+1) = \frac{3}{2}n(n+1)$

Combined Sequences

If two or more sequences are added, subtracted, multiplied or divided, the n^{th} term follows the same operation.

E.g $x = 3n + 2$, $y = (n + 1)^2$, $T_n = x + y = 3n + 2 + (n + 1)^2 = n^2 + 5n + 3$

Quadratic Sequences

If the differences are not constant, but the difference of first differences is constant, it calls for Quadratic Sequence. There are 2 methods for n^{th} term formulas.

$T_n = an^2 + bn + c$, which a is half of constant difference $\left(\frac{d_2 - d_1}{2}\right)$, and for b and c employ the method:

1. Subtract each term of original sequence by each term of $\frac{a}{2}$
2. Find n^{th} term formula for the answer, A_n .
3. Add A_n to $\frac{a}{2}$.

E.g Sequence is 2, 5, 9, 14, 20,

$$d_1 = T_2 - T_1 = 5 - 2 = 3, \quad d_2 = T_3 - T_2 = 9 - 5 = 4,$$

$$a = \frac{d_2 - d_1}{2} = \frac{4 - 3}{2} = \frac{1}{2},$$

T_n	2	5	9	...
$-\frac{1}{2}n^2$	0.5	2	4.5	...
A_n	1.5	3	4.5	...

$$d = 3 - 1.5 = 1.5$$

$$A_n = 1.5 + 1.5(n - 1) = 1.5n$$

$$T_n = \frac{1}{2}n^2 + \frac{3}{2}n = \frac{n^2 + 3n}{2}$$



Quadratic Sequences

$T_n = a + (n-1)d + \frac{1}{2}(n-1)(n-2)c$, which a is first term, d is first difference ($T_2 - T_1$), and c is the constant difference.

E.g Sequence is 2, 5, 9, 14, 20,

$$a = T_1 = 2,$$

$$d_1 = T_2 - T_1 = 5 - 2 = 3, \quad d_2 = T_3 - T_2 = 9 - 5 = 4,$$

$$c = d_2 - d_1 = 4 - 3 = 1$$

$$T_n = 2 + (n-1)(3) + \frac{1}{2}(n-1)(n-2)(1) = \frac{n^2 + 3n}{2}$$

Cubic Sequences

Core and Extended

If the difference between terms is not constant, and the difference of differences is not constant, but the difference of differences of differences is constant, it calls for Cubic Sequence.

$T_n = an^3 + bn^2 + cn + d$, in it we can derive formulas for a , b , c and d .

For this we need to know T_1 , T_2 , T_3 and T_4 to use.

$$\text{First term } T_1 = a + b + c + d$$

$$\text{First diff } (T_2 - T_1) = 7a + 3b + c$$

$$\text{Second diff } (T_3 - 2T_2 + T_1) = 12a + 2b$$

$$\text{Third diff } (T_4 - 3T_3 + 3T_2 - T_1) = 6a$$

E.g Sequence is 6, 19, 48, 99, 178,

$$\text{Third diff} = 99 - 3(48) + 3(19) - 6 = 6 = 6a \quad a = 1$$

$$\text{Second diff} = 48 - 2(19) + 6 = 16 = 12(1) + 2b \quad b = 2$$

$$\text{First diff} = 19 - 6 = 13 = 7(1) + 3(2) + c \quad c = 0$$

$$\text{First term} = 6 = (1) + (2) + (0) + d \quad d = 3$$

$$T_n = (1)n^3 + (2)n^2 + 0 + 3 = n^3 + 2n^2 + 3$$

Geometric series

Core and Extended

If the ratio between terms is constant, it calls for Geometric Sequence.

$T_n = ar^{n-1}$, which a is the first term (T_1), r is the ratio $\left(\frac{T_2}{T_1}\right)$, n is the term number.



Variation

Covering

- Proportion

Variation

There exists 2 types; Direct Proportion and Indirect Proportion.

Direct Proportion

”y is directionally proportional to x” means and calls for $y = kx$.

For this the values are both in the numerator(top). $y = kx$ is the principle equation.

Inverse Proportion

”y is inversely proportional to x” means and calls for $y = \frac{k}{x}$.

For this one value is in numerator(top), and other one is in the denominator(bottom).

$y = \frac{k}{x}$ is the principle equation.

Word Problems

Word problems will show variation type. E.g Work is inverse to Time taken, or men working. Speed is Inverse to Time. Speed is proportional to Distance, Cost is proportional to Amount.

Solving

To make a formula or write a relation between the variables, we first solve for k , direct proportion has $\frac{y}{x} = k$ and inverse proportion has $yx = k$. k is found by the given pair of values, either by table or a statement.

To find the value if one is some value, find k by getting principle equation, plug in given values to solve for k , then then use k and one of the given value to solve for the other variable.



Graphs in Practical Situations

Covering

- Graphs in practical situations

Units

Graphs are either with distance in metres or kilometres, and time in seconds or hours. Speed

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{\text{m}}{\text{s}} = \frac{\text{km}}{\text{hr}}$$

Distance Time Graphs

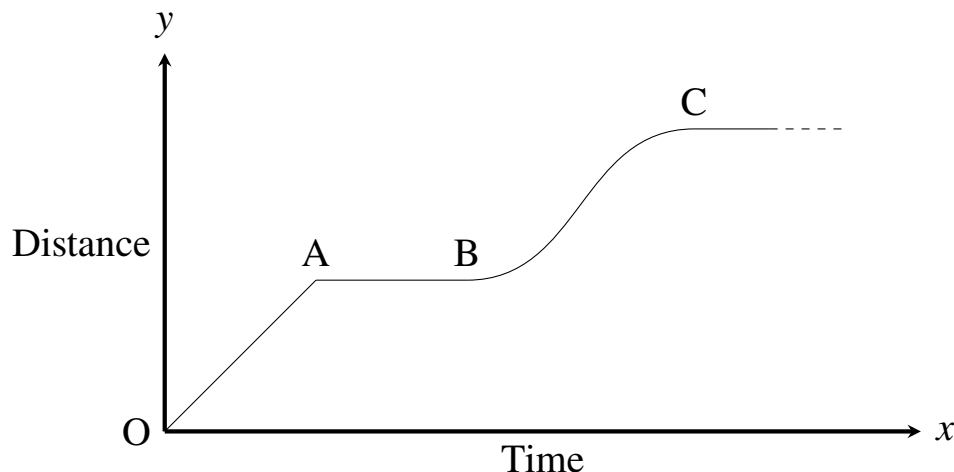
Time is on the x -axis in seconds or hours.

Distance is on the y -axis in metres or kilometres.

For each situation take the corresponding time and distance for it.

E.g In the time period stated, how much distance is covered. For the following distance, how much time it took to cover it. etc.

A distance-time graph is like so, each segment represents what is happening.



From O to A, Speed is constant. The slope is speed, i.e the gradient of the segment.

From A to B, the speed is 0, the subject is at rest.

From B to C, the speed is increasing. The curve going up right means increasing.



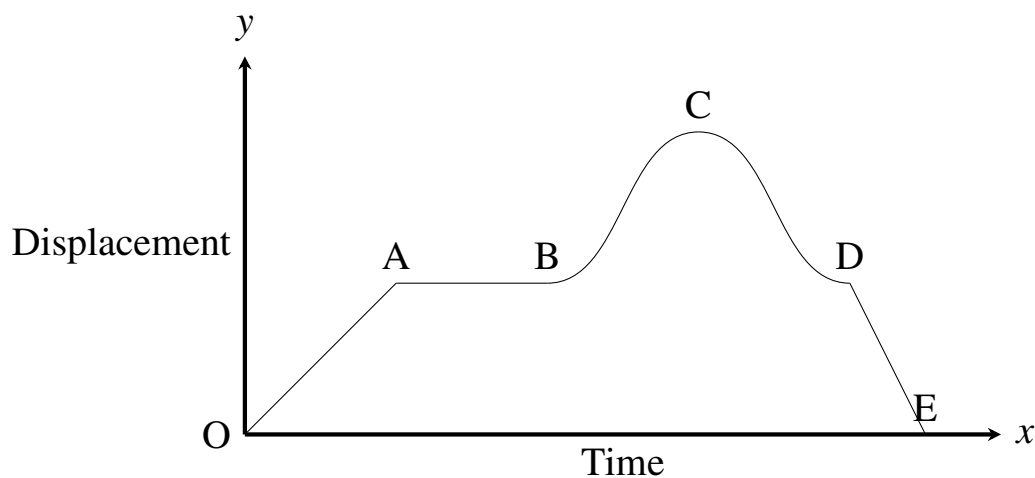
Displacement Time Graphs

Time is on the x -axis in seconds or hours.

Displacement is on the y -axis in metres or kilometres.

For each situation take the corresponding time and distance for it.

A displacement-time graph is like so, each segment represents what is happening. It is similar to distance-time graph, but it can have places which it goes down.



From O to A, and D to E, Velocity is constant. The slope is the velocity, i.e the gradient of the segment

From A to B, the velocity is 0, the subject is at rest.

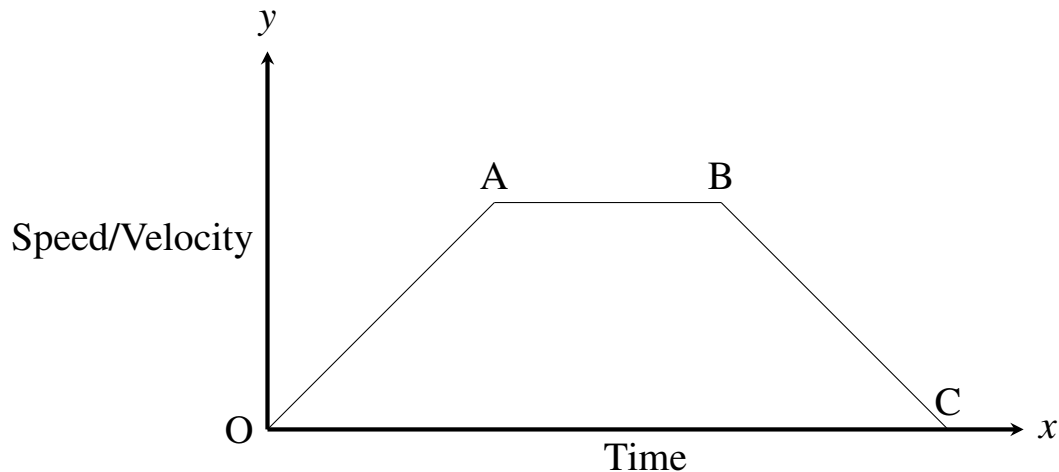
From B to C, the velocity is increasing. The curve going up right means increasing.

From C to D, the velocity is decreasing. The curve going down right means decreasing.



Speed/Velocity Time Graphs

A speed-time graph is like so, each segment represents what is happening.



From O to A, constant acceleration. The slope is the acceleration, i.e the gradient of the segment.

From A to B, constant speed. The acceleration is 0.

From B to C, constant deceleration/retardation. The slope is the deceleration, i.e i.e the gradient of the segment $\times -1$.

The Area under the graph, e.g OABC, is distance.

Speed is at times replaced by Velocity, the rate of change of distance with respect to time.
Acceleration is the rate of change of velocity with respect to time.

Speed/Velocity have the units, $\text{m/s} = \frac{\text{Distance (m)}}{\text{Time (s)}}$ or $\text{km/h} = \frac{\text{Distance (km)}}{\text{Time (h)}}$

Acceleration has the unit $\text{m/s}^2 = \frac{\text{Speed/Velocity (m/s)}}{\text{Time (s)}}$.



Graphs of Functions

Covering

- Graphs of functions
- Sketching curves

Unknown Values

For this we plug in the value of x without its y value and evaluate it to find it.

Tangent

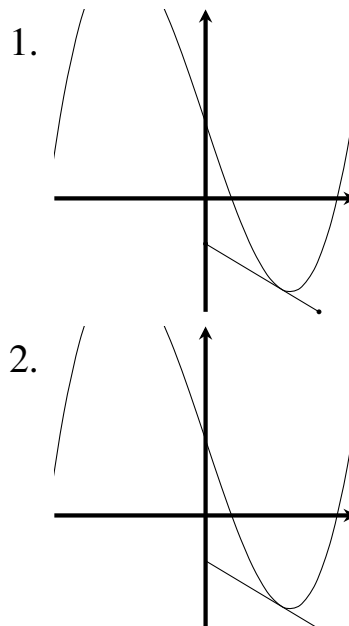
O levels

For this, 1. we get a ruler and draw a straight line approximate to how steep it is at that point. Next we sample any 2 points which we can easily see on the graph.

2. Next we use $m = \frac{y_2 - y_1}{x_2 - x_1}$ to get the answer, (x_1, y_1) as point 1 and (x_2, y_2) as point 2 coordinates.

The gradient shows "the rate of change of (variable on y -axis) with respect to (variable on x -axis)"

E.g we use points $(0, -1.2)$ and $(3, -3)$



$$m = \frac{-1.2 - (-3)}{0 - 3} = -\frac{3}{5}$$



Solution and Suitable Line

For this, if its the same equation = value, draw line $y = \text{value}$. For value = 0 its the x -axis.

If the equation is similar, then rearrange the equation to get the original equation on the left. The right side will be the equation to draw.

E.g We got $y = 3x^2 + 2x - 3$ as original equation, and it asks for solutions of $3x^2 + 2x + 1 = 0$. We solve for the original equation and see:

$$\begin{array}{rcl} 3x^2 + 2x + 1 & = & 0 \\ -4 & = & -4 \\ \hline 3x^2 + 2x - 3 & = & -4 \end{array}$$

We graph $y = -4$ on $y = 3x^2 + 2x - 3$ to find the answers for $3x^2 + 2x + 1 = 0$

For 2 equations that are equal to y , set them equal, simplify to find the intersection points or in the format it asks for.

Drawing the graph

To draw the graph, plot the values from the table and connect them in a smooth form.

Graphs of Functions, and Sketching

For these we need to know how the graphs look, the explanations of what factors effect how the graphs look are before the graphs.



Line graph

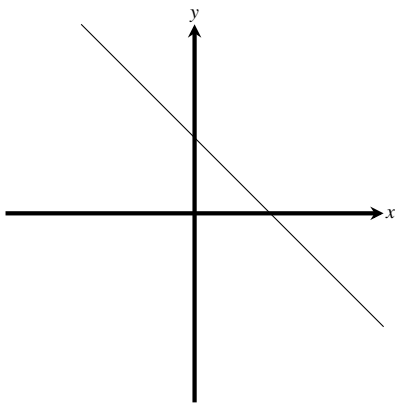
They come of the general equation $ax + by = c$. It is a straight line. After rewriting it in the form $y = mx + c$, the constant term is the y -intercept. The x -intercept is the solutions. The sign of x 's coefficient means the graph is down(-) or up (+).

To draw, we first draw the general shape. y -intercept depends on the constant value. If its positive, then it will be above the x -axis, if its negative, then it will be below the y -axis, if its not given/it is 0, it will pass through $(0,0)$.

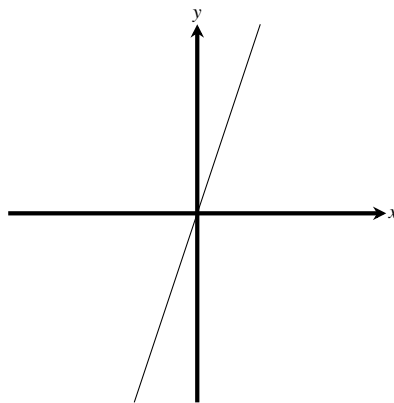
For the x -intercept, we set $y = 0$ and solve for x . Then we mark the solution as the x -intercept.

E.g we have the cases of a positive intercept with negative x , no/0 intercept and positive x , and negative intercept with positive x .

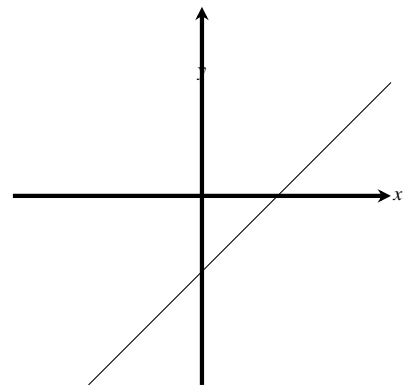
$$y = 3 - x$$



$$y = 9x$$



$$y = x - 3$$



Quadratic Graphs

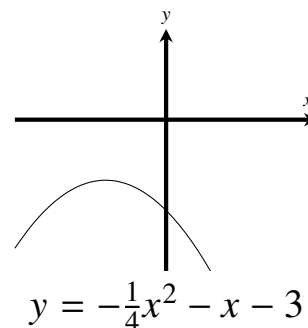
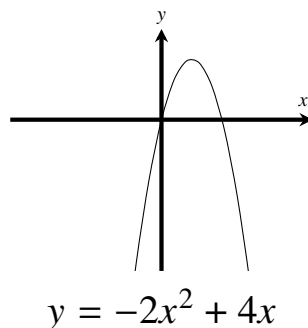
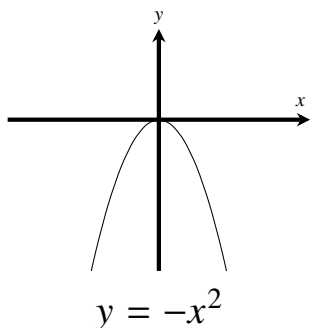
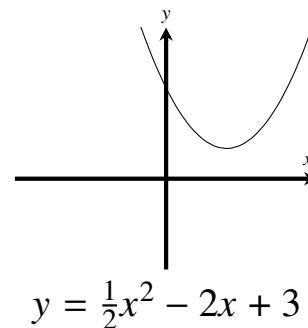
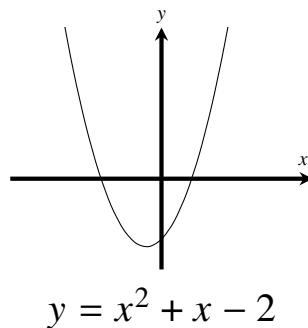
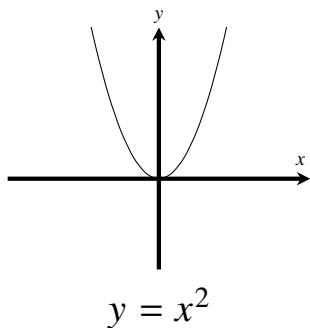
They come of the general equation $y = ax^2 + bx + c$. The constant term is the y -intercept. Any x term means the graph will have its "curve" off to some side. The sign of x^2 coefficient means the graph is happy(+) or sad(-). The x -intercept is the solutions. If it doesn't have x -intercept, it has no solution.

To draw, we first mark the intercepts. y -intercept depends on the constant value and the x term. If both aren't given/it is 0, it will pass through (0,0). y -intercept will be setting $x = 0$ and solving for y .

For the x -intercept, we set $y = 0$ and solve for x . Then we mark the solutions/roots as the x -intercept.

To calculate the "turning point", the point at which it starts to turn, "curve", it is found by $x = -\frac{b}{2a}$. If we draw a vertical line at that point that is the "line of symmetry" for the curve.

E.g we have 6 cases. Upper 3 have positive x^2 with 0 intercept, negative intercept and positive intercept. The last 2 have a x term meaning their "curves" are not on the y -axis. Lower 3 have negative x^2 with 0 intercept, 0 intercept, and negative intercept.



Reciprocal Graphs

O levels and Extended

They have the x in the denominator, they have a horizontal and vertical "asymptotes". An asymptote is where the function is not defined at that x value (vertical) or does not have a solution for $y = a$ (horizontal).

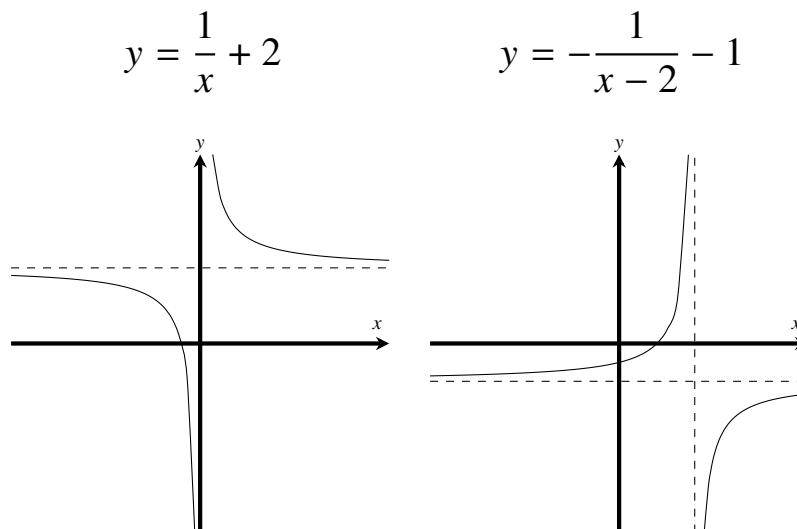
Linear Reciprocal Graphs

O levels and Extended

They come of the general equation $y = \frac{a}{x - c} + b$. The constant term is the horizontal asymptote, i.e the line where the function doesn't equal b . The first graph we see the graph never hits $y = 2$. The denominator being $x - c$ means the vertical asymptote is c . The second graph we see the graph never hits $x = 2$. The y -intercept doesn't exist for when $c = 0$, which is when the vertical asymptote is at $x = 0$. The first graph we see the graph never hits $x = 0$, which is where y -intercepts are.

To draw, we first mark the intercepts. y -intercept depends on the constant value and the value of c . y -intercept will be setting $x = 0$ and solving for y .

For the x -intercept, we set $y = 0$ and solve for x . Then we mark the solution as the x -intercept.



The dashed line is to show the horizontal and vertical asymptotes, whereas it might **not** be shown.

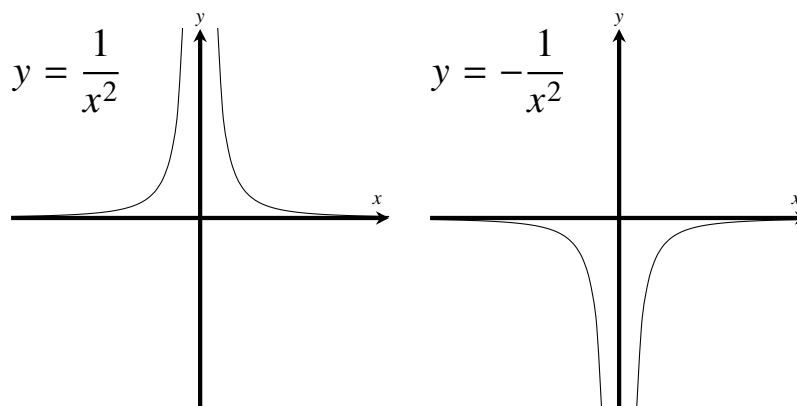


Quadratic Reciprocal Graphs

O levels and Extended

They come of the general form $y = \frac{a}{(x - c)^2} + b$. The denominator is a quadratic. Similar to Linear, else its all up. Constant term is horizontal asymptote and the vertical asymptote is c .

To draw, we first mark the intercepts. y -intercept doesn't exist because it never hits $x = 0$. For the x -intercept, we set $y = 0$ and solve for x . Then we mark the solutions/roots as the x -intercept



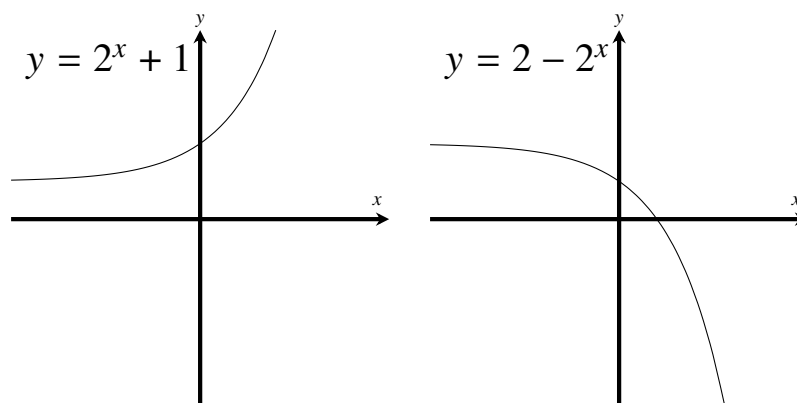
Exponential Graphs

O levels and Extended

They come of the general for $y = a^x + b$. They constant term is the horizontal asymptote also as a^x never is 0 ($a \neq 0$).

To draw, we first mark the intercepts. If $b = 0$, then the y -intercept is always 1, else y -intercept is $b + 1$ or $b - 1$, depending on coefficient of a^x .

For the x -intercept, we set $y = 0$ and solve for x . Then we mark the solution as the x -intercept.



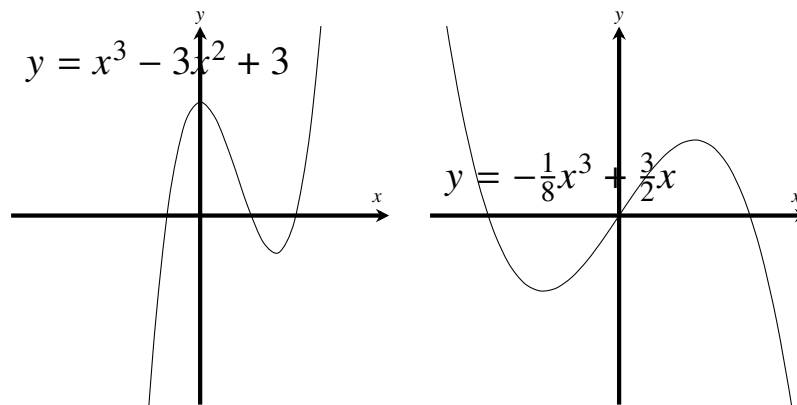
Cubic Graphs**O levels and Extended**

They come in the general form, $y = ax^3 + bx^2 + cx + d$. The constant term is the y-intercept. The sign of x^3 's coefficient shows if the graph is like a "N" or a "∩". Presence of x or x^2 terms means the graph is more or less "curvy" or to the side.

To draw, we first mark the intercepts. y-intercept depends on the constant value. If it's not given/it is 0, it will pass through (0,0).

For the x-intercept, we set $y = 0$ and solve for x . Then we mark the solutions/roots as the x-intercept.

The "line of symmetry" of the cubic graph is at $x = -\frac{b}{3a}$.



Functions

Covering

- Functions

Function

A relationship between two sets is called a function. A function is also called a mapping. In $f : A \mapsto B$ means all elements of set A is mapped to elements of set B by function f by some rule.

E.g for $f(x) = \frac{5x+3}{2x}$, f maps x to a y value by the rule $\frac{5x+3}{2x}$

The range of values that can be the input and accepted by f is the Domain of the function f .

The range of values that is output is the Range of the function f .

E.g the domain of $f(x) = \frac{5x+3}{2x}$ is $x \in \mathbb{R}$, $x \neq 0$ because $2x$ can't be 0 as its in the denominator. Similarly the range of $f(x)$ is $f(x) \in \mathbb{R}$, $f(x) \neq 2.5$ because it has a horizontal asymptote at $y = 2.5$.

For $f(x)$, domain is as x , e.g $-2 \leq x < 2$, and range is as $f(x)$, e.g $5 < f(x)$.

If the functions are tied to variables, e.g $y = f(x)$, then we may use the variable e.g we can do $5 < y$ instead of $5 < f(x)$.

Input Functions

Functions are read as: $f(x)$ "f of x". Similarly when x is replaced by a value, its

$f(4)$ "f of 4". Here you input 4 for x and solve it. $f(4) = \frac{5(4)+3}{2(4)} = \frac{23}{8}$

E.g $f(x) = 2x + 3$, $f(5) = 2(5) + 3 = 13$

Inverse Function

The inverse of a function is the reverse of that function. f^{-1} remaps some or all values of y back to x , $f^{-1} : y \mapsto x$. The Domain and Range for f^{-1} are swapped.



To find the inverse:

For the function; $f(x) = \frac{5x+3}{2x}$

1. Replace $f(x)$ and x with x and y $x = \frac{5y+3}{2y}$
2. Solve for y $x = \frac{5y+3}{2y}$
 $y = \frac{3}{2x-5}$
3. Set it equal to f^{-1} and done. $y = f^{-1} = \frac{3}{2x-5}$

For $f^{-1}(x)$, domain is as x , e.g $x \in \mathbb{R}, x \neq 0$, range is as $f^{-1}(x)$, e.g $-3 \leq f^{-1}(x) < 7$

Composite Functions

Composite functions are functions with 2 or more functions' mappings taking place. They are shown as multiple letters e.g $fg(x)$ or nested $f(g(x))$.

E.g for $f(x) = 2x + 3$ and $g(x) = 3x - 5$, we can make a few composite functions.

$fg(x)$ or $f(g(x))$, we replace $g(x)$ with its equation, $f(3x - 5)$, then we put that equation in place of x , $fg(x) = 2(3x - 5) + 3 = 6x - 7$.

$gf(x)$ or $g(f(x))$, we replace $f(x)$ with its equation, $g(2x + 3)$, then we put that equation in place of x , $gf(x) = 3(2x + 3) - 5 = 6x + 4$.

$ff(x)$ or $f(f(x))$, we replace $f(x)$ with its equation, $f(2x + 3)$, then we put that equation in place of x , $ff(x) = 2(2x + 3) + 3 = 6x + 9$. $ff(x)$ is also said as $f^2(x)$

We observe that $fg(x) \neq gf(x)$, hence the order of functions are done from inside to outside, or right to left.

We can also see it by $fg(3) = 6(3) - 7 = 2$, $gf(3) = 6(3) + 4 = 13$. $2 \neq 13$

The composite of f and f^{-1} is x . $f^{-1}(f(x)) = f^{-1}(f(x)) = x$. This fact is useful when the function is an exponential function e.g $g(x) = 5^x$. $g^{-1}(x) = 4 \rightarrow x = g(4) = 625$



Differentiation

Covering

- Differentiation

Extended

Derivatives

Derivates of a function tells us the slope of the tangent of the curve for a value of x .

The notation for a derivative is $\frac{d}{dx}$, the derivative operator. We apply this to equations, e.g

$y = ax^n$ we get $\frac{dy}{dx} = anx^{n-1}$, or functions, e.g $f(x) = 3x^2 + 6$ we get $\frac{df}{dx} = 6x$.

Gradient

We can find the gradient of the curve at some value of x by 1. differentiating the equation, 2. then plugging in the desired value of x .

E.g Find the gradient of the curve $y = x^3 - x$ at $x = 2$.

$$\begin{aligned} y &= x^3 - x \\ \frac{dy}{dx} &= 3x^2 - 1 \\ \frac{dy}{dx}(2) &= 3(2)^2 - 1 = 11 \end{aligned}$$

The second equation, $\frac{dy}{dx}$ is the equation of the tangent. In the syllabus we only differentiate terms ax^n . These are chained to have a linear, quadratic or cubic.

$$\begin{aligned} \frac{d}{dx}(ax + b) &= a \\ \frac{d}{dx}(ax^2 + bx + c) &= 2ax + b \\ \frac{d}{dx}(ax^3 + bx^2 + cx + d) &= 3ax^2 + 2bx + c \end{aligned}$$

We observe that the constant terms become 0.



Stationary points / Turning points

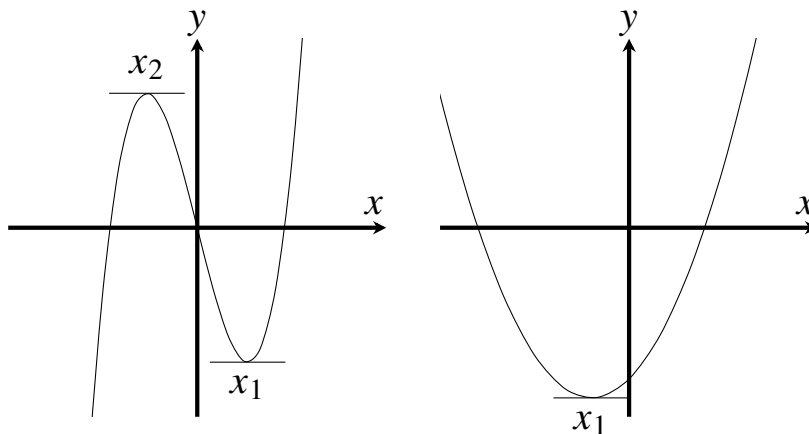
We can easily calculate the stationary points / turning points of a curve by setting the derivative to 0, and then finding the y coordinate at those values of x .

E.g Find the coordinates of the stationary points of $y = 3x^3 - 4x$

$$\frac{dy}{dx} = 9x^2 - 4 = 0, \quad x = \pm \frac{2}{3} \quad y\left(\pm \frac{2}{3}\right) = \mp \frac{16}{9} \quad \text{Therefore } \begin{matrix} x_1 = (0.667, -1.778) \\ x_2 = (-0.667, 1.778) \end{matrix}$$

E.g Find the coordinate of the stationary points of $y = x^2 + x - 2$

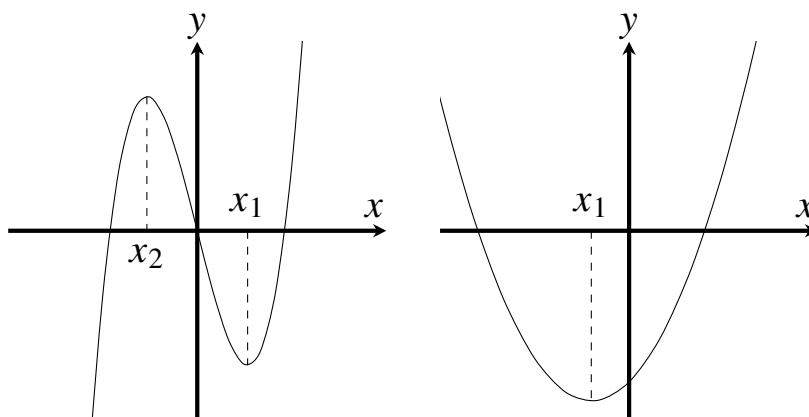
$$\frac{dy}{dx} = 2x + 1, \quad x = -\frac{1}{2} \quad y\left(-\frac{1}{2}\right) = -\frac{9}{4} \quad \text{Therefore } x_1 = (-0.5, -2.25)$$



Nature of stationary points

We can see if a point is a maxima or a minima by 3 methods:

If the graph sketch is given, then the highest point of the graph is the "maxima", and the lowest point of the graph is the "minima". The x values are found by drawing a perpendicular to the x -axis and then reading the value.

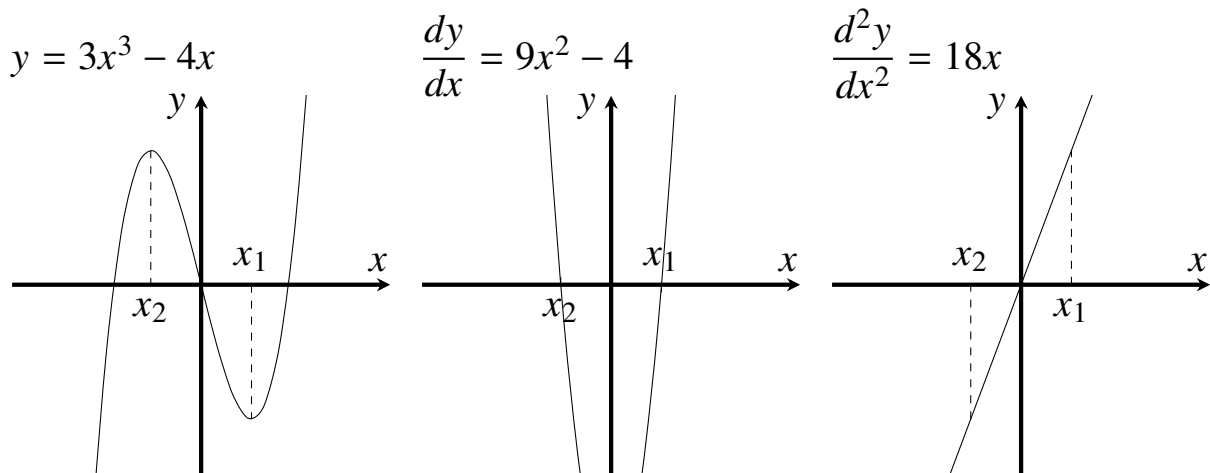


Another method is to use the second derivative of the function, $\frac{d^2y}{dx^2}$. We do this by first finding the x -coordinates of the turning points, then input them into the second derivative. If the value output is negative, its the "maximum" point. If the value output is positive, its the "minimum" point.

E.g in the cubic curve above, $\frac{d^2y}{dx^2} = 18x$.

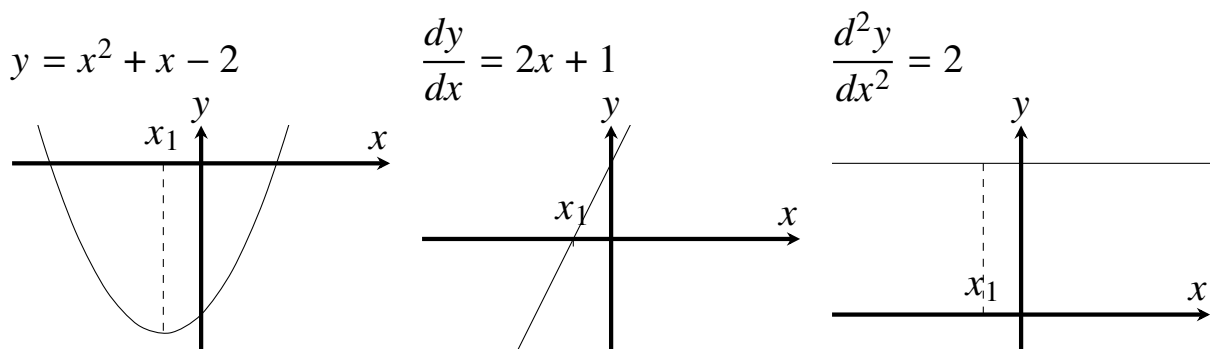
$\frac{d^2y}{dx^2} \left(\frac{2}{3} \right) = 12$ is positive, hence $(0.667, -1.778)$ is the "minimum" point.

$\frac{d^2y}{dx^2} \left(-\frac{2}{3} \right) = -12$ is negative, hence $(-0.667, 1.778)$ is the "maximum" point.



E.g in the quadratic curve above, $\frac{d^2y}{dx^2} = 2$.

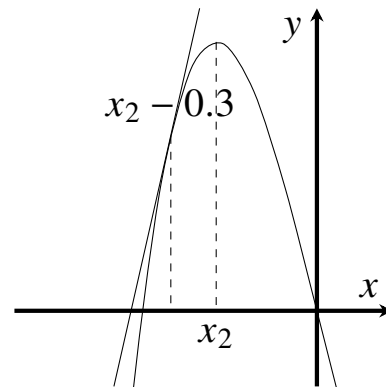
$\frac{d^2y}{dx^2} \left(-\frac{1}{2} \right) = 2$ is positive, hence $(-0.5, -2.25)$ is the "minimum point".



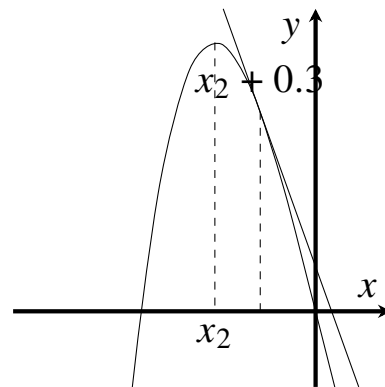
Third method is see the gradients to the left or right of the point. We do this by first finding the x -coordinates of the turning points. We only need to test one point, as the second turning point (cubic) will be the latter. For the x -value, 1. we find $\frac{d}{dx}(x - a)$ (to the left) and 2. $\frac{d}{dx}(x + a)$ (to the right) and check the signs, 3. if we get left $-$, right $+$, that point is minimum, if we get left $+$, right $-$, that point is maximum. a can be 1, 0.1, 0.5 or something easy.

E.g we will do it with the cubic, x_2 .

$$1. \frac{d}{dx} \left(-\frac{2}{3} - 0.3 \right) = 4.41, \quad +$$



$$2. \frac{d}{dx} \left(-\frac{2}{3} + 0.3 \right) = -2.79, \quad -$$



3. We see we get $+$ and $-$, meaning x_2 is a maximum. This makes x_1 a minimum.

